Physical CUME
Phase Equilibria

(1/29/2016)

1. Consider the phase change

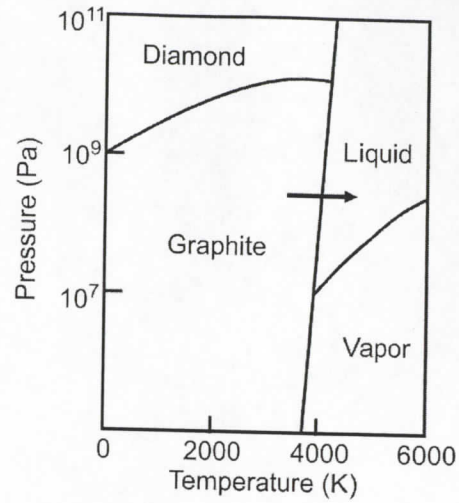
$$\text{C(graphite)} \rightleftharpoons \text{C(diamond)}$$

Given that $\Delta_r G^o/\text{J}\cdot\text{mol}^{-1} = 1895 + 3.363T$,

(a) calculate $\Delta_r H^o$ and $\Delta_r S^o$. (10 pts)

(b) Calculate the pressure at which diamond and graphite are in equilibrium with each other at 25 °C. Take the density of diamond and graphite to be 3.51 $\text{g}\cdot\text{cm}^{-3}$ and 2.25 $\text{g}\cdot\text{cm}^{-3}$, respectively. Assume that both diamond and graphite are incompressible. (25 pts)

(c) Using the P-T phase diagram of carbon (right), answer the following questions with explanation (15 pts):
i) which substance is denser, graphite or diamond?
ii) which phase is denser, graphite or liquid carbon?
iii) why does the phase diagram have two triple points?



2. The normal boiling temperature of benzene is 353.24 K, and the vapor pressure of liquid benzene is 1.19×10^4 Pa at 20.0 °C. The enthalpy of fusion ($\Delta_{fus}H$) is 9.95 $\text{kJ}\cdot\text{mol}^{-1}$, and the vapor pressure of solid benzene is 137 Pa at -44.3 °C. Calculate the following

(a) enthalpy of vaporization ($\Delta_{vap}H$) and sublimation ($\Delta_{sub}H$) (20 pts)

(b) triple point temperature and pressure. (30 pts)

If needed, you may find this helpful:

$$dG = Vdp - SdT$$

$$\frac{dp}{dT} = \frac{\Delta_{trs}S}{\Delta_{trs}V}$$

1 atm = 101325 Pa